

BIRR, IrelandWMO: **039650**Lat: **53.083N**Lon: **7.883W**Elev: **236**StdP: **14.57**Time Zone: **1.00 (IRL)**Period: **98-09**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.7	29.3	24.2	18.5	31.3	26.6	20.7	32.8	25.5	46.1	23.6	47.0	3.5	190	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	73.4	62.3	70.3	61.2	67.9	60.3	64.7	69.5	63.2	67.6	61.8	66.0	7.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.8	86.4	66.9	61.4	81.9	65.3	59.9	77.8	63.9	29.8	70.1	28.7	67.8	27.7	66.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	19.3	17.2	20.9	79.3	2.1	3.2	19.4	81.6	18.2	83.4	17.0	85.2	15.5	87.5		
(5)			20.6	68.1	2.1	2.0	19.1	69.5	17.8	70.7	16.7	71.8	15.1	73.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	42.4	42.2	44.5	47.6	52.5	56.9	59.7	59.6	56.3	50.5	45.8	42.4	
	DBStd	7.97	5.29	4.83	4.76	4.58	4.12	4.25	3.74	3.42	4.17	4.95	4.84	5.51	
	HDD50	1192	238	221	179	98	20	2	0	0	4	54	139	237	
	HDD65	5455	701	639	636	521	388	243	172	172	261	448	575	699	
	CDD50	1224	2	2	8	26	97	210	300	297	194	71	14	3	
	CDD65	14	0	0	0	0	0	1	7	5	1	0	0	0	
	CDH74	81	0	0	0	0	0	11	47	22	1	0	0	0	
	CDH80	7	0	0	0	0	0	0	5	2	0	0	0	0	
Wind	WSAvg	8.6	10.1	9.3	9.4	9.0	8.5	7.9	7.5	7.3	7.7	8.7	8.8	9.3	
Precipitation	PrecAvg	32.5	3.1	2.2	2.4	2.2	2.3	2.4	2.4	3.1	2.9	3.2	3.0	3.2	
	PrecMax	43.8	5.9	6.7	3.9	4.7	4.8	6.3	5.7	7.0	5.8	5.7	6.0	6.5	
	PrecMin	24.3	0.4	0.1	0.5	0.5	0.2	0.3	0.5	0.2	0.1	1.3	1.0	0.9	
	PrecStd	4.1	1.4	1.3	1.0	1.0	1.2	1.3	1.1	1.5	1.4	1.3	1.1	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	56.4	60.1	65.8	72.1	75.5	79.4	76.9	71.8	63.9	58.3	56.7	
		MCWB	52.7	52.7	52.3	55.6	59.6	61.3	66.3	66.6	63.8	58.8	55.1	54.0	
	2%	DB	53.3	53.1	56.2	61.3	67.7	71.6	73.9	71.0	68.4	61.5	56.3	54.0	
		MCWB	51.0	50.1	50.6	52.8	57.6	60.7	63.1	62.5	61.7	57.4	53.8	52.0	
	5%	DB	52.0	51.5	54.2	58.4	63.9	68.5	69.8	68.5	65.6	59.7	54.8	52.4	
		MCWB	49.8	48.7	49.8	51.4	55.3	59.5	61.5	61.3	59.8	56.2	52.6	50.4	
	10%	DB	50.5	50.0	52.4	55.8	60.8	65.4	66.8	66.2	63.7	58.2	53.3	50.7	
		MCWB	48.5	47.3	48.8	50.0	53.7	57.9	59.9	59.8	58.8	55.0	51.2	48.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.0	54.0	56.6	61.1	64.6	67.2	67.9	65.6	60.0	56.1	54.3
			MCDB	54.7	54.6	57.2	63.1	68.9	70.5	76.0	75.7	69.5	62.2	57.3	55.8
2%		WB	51.4	51.0	52.5	54.5	58.8	62.1	64.5	64.7	63.0	58.4	54.4	52.4	
		MCDB	52.7	52.8	55.1	59.4	65.7	68.6	71.0	68.2	66.2	60.5	55.8	53.7	
5%		WB	50.2	49.3	50.9	52.6	56.6	60.5	62.7	63.2	61.3	57.0	53.0	50.8	
		MCDB	51.6	51.1	53.3	56.9	61.7	66.7	67.9	66.5	64.5	59.1	54.4	52.2	
10%		WB	48.7	47.6	49.1	50.8	54.9	58.8	61.3	61.4	59.6	55.5	51.6	49.0	
		MCDB	50.4	49.8	51.8	54.9	59.3	64.3	65.6	64.6	62.8	57.7	53.1	50.5	
Mean Daily Temperature Range		5% DB	MDBR	9.7	11.2	12.6	14.3	14.7	14.2	13.4	13.3	13.2	12.1	10.3	9.5
			MCDBR	10.2	11.1	13.9	17.8	19.7	19.1	18.4	16.4	14.9	12.1	9.7	10.3
	MCWBR		9.5	9.6	9.6	10.6	11.1	10.4	9.4	9.0	9.0	8.8	8.8	9.7	
	5% WB	MCDBR	10.3	10.8	11.3	14.8	17.4	17.0	15.7	13.3	12.8	11.3	9.4	10.3	
		MCWBR	9.8	10.1	9.2	9.8	10.7	10.2	9.2	8.8	9.6	9.6	9.0	10.0	
Clear-Sky Solar Irradiance	taub	0.337	0.346	0.369	0.377	0.388	0.375	0.379	0.360	0.355	0.343	0.330	0.327		
	taud	2.245	2.301	2.270	2.279	2.285	2.352	2.348	2.411	2.424	2.410	2.303	2.210		
	Ebn at Noon	195	237	257	271	273	278	274	274	260	237	198	172		
	Edh at Noon	21	27	33	37	39	37	36	32	29	23	19	18		
All-Sky Solar Radiation	RadAvg	225	443	792	1225	1534	1606	1453	1228	909	532	280	171		
	RadStd	22	37	80	119	108	150	124	98	57	45	27	22		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page